MCX Natural Gas Margin & Futures Intelligence Report

Prepared for Quantitative Trading & Risk Management
Analysis Date: 03-Jun-2026

Executive Summary & Analytical Rationale

The Continuous Stitching Fallacy vs. Tradeable Futures Contracts

In quantitative research, it is common to construct a continuous price series by "stitching" the front-month contract close prices. While this is helpful for long-term charting, **it is financially and logically incorrect for evaluating directional trading swings**.

1. The Roll Yield Distortion: In physical commodities like Natural Gas, the term structure is frequently in steep contango (longer months trading at a premium due to storage and carrying costs) or backwardation. When a contract expires and the position is rolled to the next month, a trader buys the new contract at a premium or discount. This "roll gap" is not a tradeable price move of a single contract position.

2. Inherent Untradeability: A trader cannot buy a stitched index in January and hold it until December. They must buy a specific contract (e.g., January), sell it prior to expiry, buy the February contract, and so on. Over a year, the actual P&L is heavily dominated by roll yields, not the raw price change of the continuous index.

3. Margin Integrity: Initial margins (SPAN) are set on *specific contracts* by the clearing corporation. The relationship between margin changes and price moves can only be accurately measured by evaluating a contract's price path during its active, liquid life.

Therefore, this report implements a **contract-specific swing analysis** where price swings and margin parameters are tracked on individual contracts during their active liquid window (defined as trading days where daily volume is ≥ 500 lots).

TASK 1 - RECENT MARGIN ANALYSIS

The following metrics are calculated for the front-month contract (`NATURALGAS`) over the last 90 trading days in `margins.db` (covering the period **15-Jan-2026** to **03-Jun-2026**):

Key Margin Statistics (Last 90 Days)

- **Current Front-Month Initial Margin (SPAN Risk)**: **15.84%** (as of **03-Jun-2026**, for contract expiry **25-Jun-2026**).
- **30-Day Average**: **16.38%**
- **60-Day Average**: **20.85%**
- **90-Day Average**: **28.72%**
- **Margin Trend**: **Falling** (declining). The daily linear slope computed over the last 90 trading days is **-0.47%** per day.
- **Extrema**:
- *Minimum Margin %*: **14.35%** (seen on **27-May-2026**)
- *Maximum Margin %*: **66.05%** (seen on **02-Feb-2026**)

Single-Day Margin Hikes (> 2.00 pp)

- **30-Jan-2026**: Margin increased from **31.15%** to **34.86%** (**+3.71 pp**).
- **02-Feb-2026**: Margin spiked from **34.86%** to **66.05%** (**+31.19 pp**), marking a massive volatility correction by the clearing corporation.

Historical Comparison (Same Calendar Week)

The table below compares the current margin level to the same calendar week (+/- 7 days) in prior years:

Year	Average Initial Margin %	Status Relative to Current
2026 (Current)	15.84%	Baseline
2025	23.63%	Elevated (+7.79 pp)
2024	19.66%	Elevated (+3.82 pp)
2023	19.59%	Elevated (+3.75 pp)
2022	25.45%	Elevated (+9.61 pp)
2021	12.04%	Suppressed (-3.80 pp)

Historical Percentile: The current margin level of **15.84%** is at the **64.66th percentile** of all historical observations.

Margin Term Structure (as of 03-Jun-2026)

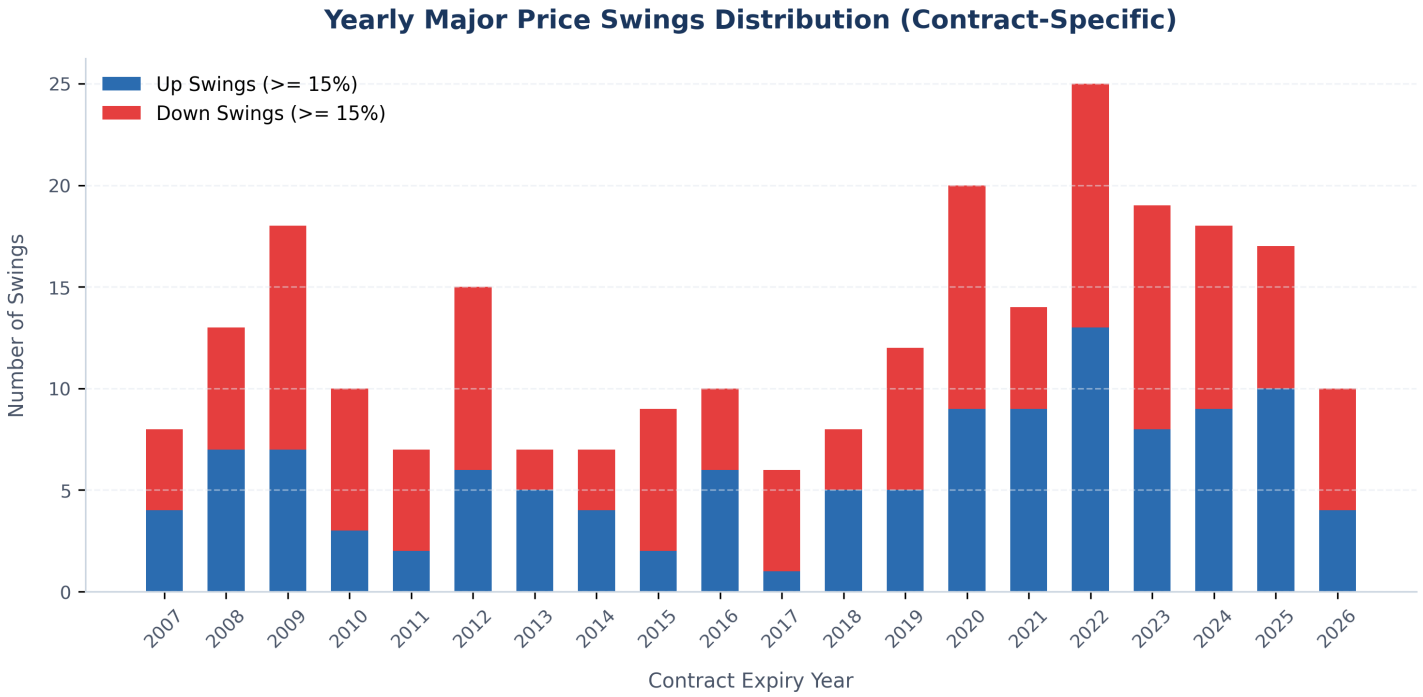
Expiry	DTE	Initial Margin %	Total Margin %	Structure State
25-Jun-2026 (Jul 2026)	**22**	**15.84%**	**17.09%**	**Front-Month**
28-Jul-2026 (Aug 2026)	**55**	**15.54%**	**16.79%**	**Backwardation (vs June)**
26-Aug-2026 (Sep 2026)	**84**	**13.97%**	**15.22%**	**Backwardation (vs July)**
25-Sep-2026 (Oct 2026)	**114**	**18.60%**	**19.85%**	**Contango (vs August)**
27-Oct-2026 (Nov 2026)	**146**	**17.17%**	**18.42%**	**Backwardation (vs September)**
24-Nov-2026 (Dec 2026)	**174**	**29.67%**	**30.92%**	**Steep Contango (vs October)**

TASK 2 - PRICE MOVEMENT ANALYSIS (YEAR BY YEAR)

Using the individual contracts' price series during their tradeable window (strictly defined by DTE range: **5 <= DTE <= 60 days**) , major moves are defined as a sustained swing of **$\ge 15.00\%$** (up or down) from local trough to peak (or peak to trough) confirmed over **≥ 5 trading days**.

Annual Performance Summary Table (Contract Swings)

Year	Largest Up Move %	Up Duration	Largest Down Move %	Down Duration	Moves >= 15%	Moves >= 20%	Moves >= 25%	Moves >= 30%
2007	23.00%	16 days	-25.49%	12 days	8	5	1	0
2008	35.34%	37 days	-40.85%	27 days	13	9	6	3
2009	43.42%	12 days	-39.88%	22 days	18	14	10	6
2010	30.95%	17 days	-28.58%	33 days	10	6	3	2
2011	15.69%	18 days	-20.10%	20 days	7	1	0	0
2012	38.64%	22 days	-38.63%	37 days	15	9	6	4
2013	26.66%	17 days	-17.14%	23 days	7	4	2	0
2014	53.74%	30 days	-22.12%	23 days	7	5	1	1
2015	20.33%	25 days	-33.92%	39 days	9	5	2	2
2016	33.39%	18 days	-29.75%	25 days	10	7	6	2
2017	15.37%	29 days	-25.81%	37 days	6	3	1	0
2018	54.53%	33 days	-27.97%	24 days	8	7	3	2
2019	28.76%	36 days	-36.52%	20 days	12	6	3	1
2020	43.30%	23 days	-30.27%	28 days	20	12	4	3
2021	47.64%	27 days	-36.60%	23 days	14	9	6	5
2022	76.35%	35 days	-45.67%	40 days	25	20	17	11
2023	31.83%	9 days	-54.82%	39 days	19	11	6	5
2024	45.92%	19 days	-42.91%	29 days	18	12	7	5
2025	47.03%	27 days	-30.54%	28 days	17	11	10	5
2026	65.23%	5 days	-37.44%	30 days	10	6	5	4



Detailed Contract Leg Dynamics (2007-2026)

To provide deeper execution intelligence for futures traders, the table below dissects every single price swing of >= 15.00% across individual contracts since 2007. It adds critical trading parameters:

- **Margin Min / Max / Avg**: The SPAN initial margin ranges during the exact duration of the swing (available post-2010).
- **Ann. Vol %**: The annualized realized price volatility of the contract close returns during the swing leg.
- **OI Change %**: The percentage change in contract open interest from the start to the end of the swing (measuring long/short positioning builds).
- **Leverage Efficiency**: Calculated as Price Move % / Average Margin %. This indicates the return on margin capital (capital efficiency multiplier) achieved by catching the swing.

Detailed Contract Leg Dynamics (2007-2026)

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2007 (20FEB2007 / Mar 2007)	UP	18-Jan-2007 to 09-Feb-2007	23.00%	N/A / N/A	N/A / N/A / N/A	50.42%	-15.98%	12069.25	N/A
2007 (20JUL2007 / Aug 2007)	DOWN	21-May-2007 to 06-Jul-2007	-22.95%	N/A / N/A	N/A / N/A / N/A	24.69%	627.55%	1048.34	N/A
2007 (20AUG2007 / Sep 2007)	DOWN	21-Jun-2007 to 24-Jul-2007	-21.80%	N/A / N/A	N/A / N/A / N/A	36.03%	593.50%	1738.17	N/A
2007 (20AUG2007 / Sep 2007)	UP	24-Jul-2007 to 14-Aug-2007	16.18%	N/A / N/A	N/A / N/A / N/A	45.43%	-53.67%	6507.38	N/A
2007 (20SEP2007 / Oct 2007)	UP	24-Jul-2007 to 17-Aug-2007	18.56%	N/A / N/A	N/A / N/A / N/A	40.29%	-15.23%	1284.17	N/A
2007 (20SEP2007 / Oct 2007)	DOWN	17-Aug-2007 to 03-Sep-2007	-25.49%	N/A / N/A	N/A / N/A / N/A	62.17%	790.66%	6797.75	N/A
2007 (20SEP2007 / Oct 2007)	UP	03-Sep-2007 to 14-Sep-2007	16.67%	N/A / N/A	N/A / N/A / N/A	48.84%	-72.12%	11057.50	N/A
2007 (20DEC2007 / Jan 2008)	DOWN	01-Nov-2007 to 14-Dec-2007	-20.50%	N/A / N/A	N/A / N/A / N/A	35.90%	104.26%	1372.97	N/A
2008 (18JAN2008 / Feb 2008)	DOWN	19-Nov-2007 to 27-Dec-2007	-15.29%	N/A / N/A	N/A / N/A / N/A	30.04%	991.36%	501.29	N/A
2008 (18JAN2008 / Feb 2008)	UP	27-Dec-2007 to 10-Jan-2008	16.18%	N/A / N/A	N/A / N/A / N/A	26.47%	-54.07%	2814.55	N/A
2008 (20FEB2008 / Mar 2008)	UP	27-Dec-2007 to 15-Feb-2008	22.18%	N/A / N/A	N/A / N/A / N/A	26.27%	223.02%	1011.43	N/A
2008 (20MAR2008 / Apr 2008)	UP	23-Jan-2008 to 13-Mar-2008	35.34%	N/A / N/A	N/A / N/A / N/A	26.40%	35.11%	1420.78	N/A
2008 (18APR2008 / May 2008)	UP	18-Feb-2008 to 14-Mar-2008	18.74%	N/A / N/A	N/A / N/A / N/A	24.19%	2268.29%	565.15	N/A
2008 (20MAY2008 / Jun 2008)	UP	24-Mar-2008 to 14-May-2008	32.63%	N/A / N/A	N/A / N/A / N/A	29.76%	1238.89%	771.16	N/A
2008 (20JUN2008 / Jul 2008)	UP	21-Apr-2008 to 12-Jun-2008	27.66%	N/A / N/A	N/A / N/A / N/A	27.77%	3065.00%	1028.97	N/A
2008 (18JUL2008 / Aug 2008)	UP	19-May-2008 to 01-Jul-2008	21.87%	N/A / N/A	N/A / N/A / N/A	25.44%	5911.11%	589.47	N/A
2008 (20AUG2008 / Sep 2008)	DOWN	04-Jul-2008 to 11-Aug-2008	-40.85%	N/A / N/A	N/A / N/A / N/A	42.98%	1645.00%	2471.56	N/A
2008 (19SEP2008 / Oct 2008)	DOWN	21-Jul-2008 to 04-Sep-2008	-29.32%	N/A / N/A	N/A / N/A / N/A	53.15%	24042.86%	1708.73	N/A
2008 (20OCT2008 / Nov 2008)	DOWN	27-Aug-2008 to 13-Oct-2008	-16.25%	N/A / N/A	N/A / N/A / N/A	44.62%	831.25%	1817.42	N/A
2008 (20NOV2008 / Dec 2008)	DOWN	23-Sep-2008 to 14-Nov-2008	-20.80%	N/A / N/A	N/A / N/A / N/A	45.59%	2006.25%	855.50	N/A
2008 (19DEC2008 / Jan 2009)	DOWN	20-Oct-2008 to 12-Dec-2008	-26.92%	N/A / N/A	N/A / N/A / N/A	46.90%	5823.08%	1468.90	N/A
2009 (20JAN2009 / Feb 2009)	DOWN	26-Nov-2008 to 15-Jan-2009	-31.22%	N/A / N/A	N/A / N/A / N/A	52.68%	4975.00%	1933.14	N/A
2009 (20FEB2009 / Mar 2009)	UP	22-Dec-2008 to 29-Dec-2008	15.10%	N/A / N/A	N/A / N/A / N/A	62.70%	129.17%	78.80	N/A
2009 (20FEB2009 / Mar 2009)	DOWN	29-Dec-2008 to 28-Jan-2009	-26.71%	N/A / N/A	N/A / N/A / N/A	44.44%	3798.18%	1281.23	N/A
2009 (20MAR2009 / Apr 2009)	DOWN	21-Jan-2009 to 11-Mar-2009	-16.36%	N/A / N/A	N/A / N/A / N/A	35.35%	10618.75%	2779.63	N/A

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2009 (20APR2009 / May 2009)	DOWN	03-Mar-2009 to 09-Apr-2009	-20.06%	N/A / N/A	N/A / N/A / N/A	46.92%	9737.50%	7388.46	N/A
2009 (20MAY2009 / Jun 2009)	DOWN	24-Mar-2009 to 27-Apr-2009	-26.72%	N/A / N/A	N/A / N/A / N/A	39.39%	12327.52%	4934.25	N/A
2009 (19JUN2009 / Jul 2009)	UP	27-Apr-2009 to 13-May-2009	27.82%	N/A / N/A	N/A / N/A / N/A	38.55%	71.03%	2123.58	N/A
2009 (20MAY2009 / Jun 2009)	UP	27-Apr-2009 to 13-May-2009	30.61%	N/A / N/A	N/A / N/A / N/A	41.66%	-80.36%	24789.83	N/A
2009 (19JUN2009 / Jul 2009)	DOWN	13-May-2009 to 25-May-2009	-24.80%	N/A / N/A	N/A / N/A / N/A	39.44%	692.79%	12672.56	N/A
2009 (20JUL2009 / Aug 2009)	UP	25-May-2009 to 17-Jun-2009	18.56%	N/A / N/A	N/A / N/A / N/A	56.67%	338.64%	7705.22	N/A
2009 (20JUL2009 / Aug 2009)	DOWN	17-Jun-2009 to 13-Jul-2009	-24.85%	N/A / N/A	N/A / N/A / N/A	28.90%	375.18%	44649.84	N/A
2009 (20AUG2009 / Sep 2009)	DOWN	22-Jun-2009 to 14-Aug-2009	-22.05%	N/A / N/A	N/A / N/A / N/A	55.67%	1873.24%	35831.62	N/A
2009 (25SEP2009 / Oct 2009)	DOWN	05-Aug-2009 to 03-Sep-2009	-39.88%	N/A / N/A	N/A / N/A / N/A	37.96%	347.76%	31611.14	N/A
2009 (25SEP2009 / Oct 2009)	UP	03-Sep-2009 to 18-Sep-2009	43.42%	N/A / N/A	N/A / N/A / N/A	71.67%	-65.34%	84941.33	N/A
2009 (27OCT2009 / Nov 2009)	UP	04-Sep-2009 to 21-Oct-2009	34.31%	N/A / N/A	N/A / N/A / N/A	58.61%	1.91%	37187.73	N/A
2009 (23NOV2009 / Dec 2009)	DOWN	29-Sep-2009 to 18-Nov-2009	-27.69%	N/A / N/A	N/A / N/A / N/A	43.49%	519.03%	21583.42	N/A
2009 (28DEC2009 / Jan 2010)	DOWN	29-Oct-2009 to 03-Dec-2009	-19.17%	N/A / N/A	N/A / N/A / N/A	42.60%	2046.82%	17284.42	N/A
2009 (28DEC2009 / Jan 2010)	UP	03-Dec-2009 to 17-Dec-2009	32.48%	N/A / N/A	N/A / N/A / N/A	51.55%	-36.22%	45677.18	N/A
2010 (25JAN2010 / Feb 2010)	UP	03-Dec-2009 to 28-Dec-2009	30.95%	N/A / N/A	N/A / N/A / N/A	48.10%	229.22%	6304.35	N/A
2010 (26MAR2010 / Apr 2010)	DOWN	25-Jan-2010 to 18-Mar-2010	-28.58%	8.87% / 7.58%	6.51% / 11.37% / 8.70%	32.39%	3540.48%	9338.64	3.28x
2010 (27APR2010 / May 2010)	DOWN	26-Feb-2010 to 08-Apr-2010	-22.02%	7.66% / 8.40%	6.17% / 8.40% / 6.94%	32.36%	3801.42%	16047.88	3.17x
2010 (25JUN2010 / Jul 2010)	UP	30-Apr-2010 to 15-Jun-2010	30.21%	10.98% / 8.92%	6.83% / 10.98% / 8.79%	34.58%	558.75%	13946.91	3.44x
2010 (27JUL2010 / Aug 2010)	UP	01-Jun-2010 to 15-Jun-2010	16.56%	6.71% / 8.38%	6.71% / 9.54% / 8.26%	42.07%	20.19%	2099.64	2.00x
2010 (27JUL2010 / Aug 2010)	DOWN	15-Jun-2010 to 14-Jul-2010	-15.54%	8.38% / 9.45%	8.17% / 11.15% / 9.30%	44.48%	1171.65%	22082.50	1.67x
2010 (26AUG2010 / Sep 2010)	DOWN	01-Jul-2010 to 20-Aug-2010	-16.76%	8.01% / 6.80%	6.80% / 10.62% / 8.41%	38.36%	1655.62%	25122.22	1.99x
2010 (27SEP2010 / Oct 2010)	DOWN	30-Jul-2010 to 09-Sep-2010	-23.32%	6.45% / 6.37%	6.00% / 7.71% / 6.88%	28.43%	4699.00%	21144.00	3.39x
2010 (26OCT2010 / Nov 2010)	DOWN	31-Aug-2010 to 21-Oct-2010	-24.26%	6.65% / 9.00%	5.69% / 9.00% / 7.00%	34.29%	1419.72%	28348.61	3.47x
2010 (23NOV2010 / Dec 2010)	DOWN	24-Sep-2010 to 25-Oct-2010	-16.41%	5.56% / 5.81%	5.23% / 6.22% / 5.75%	25.01%	783.21%	5157.36	2.85x
2011 (23FEB2011 / Mar 2011)	UP	27-Dec-2010 to 21-Jan-2011	15.31%	6.20% / 6.54%	5.91% / 7.34% / 6.68%	31.29%	89.49%	6058.40	2.29x
2011 (23FEB2011 / Mar 2011)	DOWN	21-Jan-2011 to 18-Feb-2011	-20.10%	6.54% / 6.41%	6.10% / 7.21% / 6.68%	25.32%	152.68%	32843.80	3.01x
2011 (28MAR2011 / Apr 2011)	DOWN	27-Jan-2011 to 03-Mar-2011	-15.81%	7.23% / 6.78%	5.60% / 7.23% / 6.42%	26.25%	1023.71%	11910.50	2.46x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2011 (26APR2011 / May 2011)	UP	03-Mar-2011 to 28-Mar-2011	15.69%	6.35% / 5.61%	5.34% / 6.35% / 5.78%	20.36%	192.82%	4397.06	2.71x
2011 (26JUL2011 / Aug 2011)	DOWN	08-Jun-2011 to 07-Jul-2011	-16.05%	5.86% / 6.17%	5.28% / 6.72% / 6.12%	26.87%	700.94%	13723.18	2.62x
2011 (25NOV2011 / Dec 2011)	DOWN	26-Sep-2011 to 18-Nov-2011	-16.02%	5.00% / 5.00%	5.00% / 5.00% / 5.00%	19.76%	2010.50%	21912.85	3.20x
2011 (27DEC2011 / Jan 2012)	DOWN	28-Oct-2011 to 20-Dec-2011	-17.30%	5.00% / 5.72%	5.00% / 6.60% / 5.28%	24.38%	2792.27%	25748.63	3.27x
2012 (25JAN2012 / Feb 2012)	DOWN	01-Dec-2011 to 20-Jan-2012	-38.63%	5.00% / 10.07%	5.00% / 10.38% / 6.53%	34.26%	540.21%	35538.68	5.92x
2012 (24FEB2012 / Mar 2012)	DOWN	26-Dec-2011 to 01-Feb-2012	-31.20%	5.07% / 13.52%	5.00% / 13.52% / 8.19%	50.93%	2293.30%	29978.37	3.81x
2012 (27MAR2012 / Apr 2012)	DOWN	30-Jan-2012 to 13-Mar-2012	-22.55%	9.42% / 9.33%	9.09% / 12.28% / 10.35%	45.51%	1348.06%	32473.88	2.18x
2012 (25APR2012 / May 2012)	DOWN	27-Feb-2012 to 19-Apr-2012	-27.97%	8.45% / 7.38%	6.93% / 9.57% / 8.16%	33.42%	1906.03%	36352.32	3.43x
2012 (25MAY2012 / Jun 2012)	DOWN	26-Mar-2012 to 19-Apr-2012	-17.82%	6.75% / 7.05%	6.62% / 8.27% / 7.13%	28.12%	423.44%	7608.56	2.50x
2012 (25MAY2012 / Jun 2012)	UP	19-Apr-2012 to 18-May-2012	38.64%	7.05% / 11.34%	6.84% / 11.34% / 8.46%	47.88%	75.87%	84716.41	4.57x
2012 (26JUN2012 / Jul 2012)	UP	27-Apr-2012 to 23-May-2012	29.01%	6.77% / 10.08%	6.77% / 10.62% / 8.61%	46.89%	266.77%	13166.95	3.37x
2012 (26JUN2012 / Jul 2012)	DOWN	23-May-2012 to 13-Jun-2012	-22.17%	10.08% / 11.41%	10.08% / 12.55% / 11.20%	51.11%	125.97%	71938.00	1.98x
2012 (26JUL2012 / Aug 2012)	UP	13-Jun-2012 to 20-Jul-2012	32.89%	10.97% / 9.84%	9.28% / 13.95% / 11.24%	48.64%	386.89%	68158.86	2.93x
2012 (26JUN2012 / Jul 2012)	UP	13-Jun-2012 to 21-Jun-2012	19.07%	11.41% / 13.10%	11.41% / 14.52% / 13.49%	66.03%	-49.32%	99849.57	1.41x
2012 (28AUG2012 / Sep 2012)	UP	16-Jul-2012 to 24-Jul-2012	15.61%	9.43% / 9.36%	9.10% / 9.90% / 9.45%	37.35%	161.08%	19841.29	1.65x
2012 (28AUG2012 / Sep 2012)	DOWN	24-Jul-2012 to 23-Aug-2012	-15.01%	9.36% / 10.12%	9.36% / 12.61% / 10.39%	50.15%	106.68%	116390.77	1.44x
2012 (25SEP2012 / Oct 2012)	DOWN	31-Jul-2012 to 28-Aug-2012	-17.49%	9.26% / 8.55%	8.55% / 11.73% / 9.79%	40.82%	741.47%	22255.30	1.79x
2012 (26OCT2012 / Nov 2012)	UP	07-Sep-2012 to 19-Oct-2012	24.33%	7.96% / 7.97%	7.08% / 10.22% / 8.28%	40.81%	391.43%	69047.90	2.94x
2012 (26DEC2012 / Jan 2013)	DOWN	22-Nov-2012 to 14-Dec-2012	-19.27%	5.94% / 6.95%	5.76% / 7.54% / 6.83%	29.81%	95.46%	85145.47	2.82x
2013 (28JAN2013 / Feb 2013)	DOWN	06-Dec-2012 to 09-Jan-2013	-16.57%	6.88% / 8.27%	6.39% / 8.39% / 7.11%	34.71%	980.87%	56482.08	2.33x
2013 (25MAR2013 / Apr 2013)	UP	14-Feb-2013 to 19-Mar-2013	23.60%	7.03% / 5.49%	5.25% / 7.03% / 5.94%	25.42%	815.65%	93434.21	3.97x
2013 (25APR2013 / May 2013)	UP	26-Feb-2013 to 18-Apr-2013	23.58%	5.90% / 7.22%	5.00% / 7.22% / 5.62%	27.61%	1006.39%	87124.35	4.20x
2013 (25SEP2013 / Oct 2013)	UP	09-Aug-2013 to 03-Sep-2013	26.66%	5.71% / 7.46%	5.71% / 7.46% / 7.03%	31.07%	216.41%	14151.24	3.79x
2013 (27AUG2013 / Sep 2013)	UP	09-Aug-2013 to 22-Aug-2013	15.12%	5.95% / 7.33%	5.95% / 7.49% / 7.01%	31.77%	-43.11%	34540.78	2.16x
2013 (28OCT2013 / Nov 2013)	DOWN	03-Sep-2013 to 04-Oct-2013	-17.14%	7.15% / 7.51%	6.27% / 8.31% / 7.07%	31.37%	311.74%	11427.96	2.42x
2013 (26DEC2013 / Jan 2014)	UP	04-Nov-2013 to 19-Dec-2013	25.37%	6.40% / 6.61%	5.28% / 6.61% / 6.02%	26.24%	950.22%	34173.97	4.21x
2014 (28JAN2014 / Feb 2014)	UP	29-Nov-2013 to 23-Jan-2014	19.37%	5.64% / 9.04%	5.14% / 9.04% / 6.30%	33.72%	750.54%	37839.46	3.07x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2014 (25FEB2014 / Mar 2014)	UP	10-Jan-2014 to 20-Feb-2014	53.74%	6.54% / 14.97%	6.54% / 15.37% / 10.08%	57.55%	439.21%	81453.07	5.33x
2014 (28JUL2014 / Aug 2014)	DOWN	17-Jun-2014 to 23-Jul-2014	-20.93%	6.12% / 6.14%	5.24% / 6.39% / 5.95%	22.29%	305.47%	27235.81	3.52x
2014 (26AUG2014 / Sep 2014)	DOWN	30-Jun-2014 to 28-Jul-2014	-15.74%	5.69% / 5.70%	5.00% / 6.06% / 5.66%	22.25%	1128.32%	3545.24	2.78x
2014 (24NOV2014 / Dec 2014)	UP	27-Oct-2014 to 07-Nov-2014	21.85%	5.35% / 7.26%	5.35% / 7.26% / 6.06%	22.03%	6.97%	46365.50	3.60x
2014 (26DEC2014 / Jan 2015)	UP	27-Oct-2014 to 19-Nov-2014	22.38%	5.27% / 8.63%	5.27% / 8.63% / 6.68%	36.69%	129.35%	4475.94	3.35x
2014 (26DEC2014 / Jan 2015)	DOWN	19-Nov-2014 to 19-Dec-2014	-22.12%	8.63% / 10.15%	8.34% / 11.26% / 9.82%	45.27%	248.27%	54025.26	2.25x
2015 (27JAN2015 / Feb 2015)	DOWN	28-Nov-2014 to 22-Jan-2015	-33.92%	8.65% / 14.85%	8.65% / 15.53% / 11.71%	57.52%	185.43%	37105.92	2.90x
2015 (24FEB2015 / Mar 2015)	DOWN	29-Dec-2014 to 06-Feb-2015	-21.45%	11.41% / 12.59%	11.05% / 14.61% / 12.86%	57.25%	678.93%	19961.24	1.67x
2015 (27APR2015 / May 2015)	DOWN	18-Mar-2015 to 13-Apr-2015	-15.47%	9.76% / 8.36%	8.36% / 10.17% / 9.08%	30.91%	905.08%	25634.44	1.70x
2015 (26MAY2015 / Jun 2015)	UP	13-Apr-2015 to 15-May-2015	20.33%	8.03% / 8.12%	7.36% / 9.00% / 8.10%	33.80%	460.41%	26200.04	2.51x
2015 (25JUN2015 / Jul 2015)	UP	27-Apr-2015 to 15-May-2015	19.44%	7.30% / 7.87%	7.09% / 8.66% / 8.01%	35.24%	307.89%	1710.87	2.43x
2015 (25JUN2015 / Jul 2015)	DOWN	15-May-2015 to 05-Jun-2015	-15.37%	7.87% / 7.38%	7.12% / 8.45% / 7.67%	25.87%	1014.65%	23384.06	2.00x
2015 (27OCT2015 / Nov 2015)	DOWN	14-Sep-2015 to 22-Oct-2015	-18.10%	5.11% / 5.54%	5.00% / 6.50% / 5.51%	23.06%	1889.03%	29868.46	3.28x
2015 (24NOV2015 / Dec 2015)	DOWN	28-Sep-2015 to 02-Nov-2015	-21.59%	5.00% / 7.49%	5.00% / 7.72% / 5.66%	28.36%	613.56%	15966.92	3.81x
2015 (28DEC2015 / Jan 2016)	DOWN	16-Nov-2015 to 18-Dec-2015	-31.18%	7.39% / 9.14%	7.06% / 9.17% / 8.11%	31.52%	417.88%	35778.24	3.84x
2016 (25JAN2016 / Feb 2016)	DOWN	26-Nov-2015 to 18-Dec-2015	-19.39%	7.43% / 8.12%	6.72% / 8.31% / 7.48%	26.37%	138.68%	2703.12	2.59x
2016 (25JAN2016 / Feb 2016)	UP	18-Dec-2015 to 08-Jan-2016	30.70%	8.12% / 13.27%	8.12% / 14.34% / 12.07%	64.63%	51.56%	48136.00	2.54x
2016 (24FEB2016 / Mar 2016)	DOWN	08-Jan-2016 to 19-Feb-2016	-24.65%	11.44% / 11.69%	9.22% / 13.18% / 11.44%	46.62%	946.35%	28444.33	2.15x
2016 (28MAR2016 / Apr 2016)	DOWN	29-Jan-2016 to 03-Mar-2016	-29.75%	8.39% / 9.01%	8.39% / 10.26% / 9.36%	37.28%	4689.98%	16109.16	3.18x
2016 (27JUN2016 / Jul 2016)	UP	18-May-2016 to 21-Jun-2016	29.75%	7.36% / 9.11%	6.63% / 9.59% / 8.16%	37.90%	817.56%	43653.88	3.65x
2016 (26JUL2016 / Aug 2016)	UP	30-May-2016 to 01-Jul-2016	29.27%	5.90% / 7.86%	5.90% / 8.51% / 7.64%	34.65%	1669.20%	16294.12	3.83x
2016 (27SEP2016 / Oct 2016)	UP	11-Aug-2016 to 21-Sep-2016	16.84%	8.54% / 7.93%	7.55% / 8.54% / 7.99%	31.87%	732.04%	34676.72	2.11x
2016 (26OCT2016 / Nov 2016)	UP	07-Sep-2016 to 13-Oct-2016	18.99%	6.62% / 7.97%	6.23% / 7.97% / 6.84%	31.60%	1006.38%	26611.56	2.78x
2016 (25NOV2016 / Dec 2016)	DOWN	13-Oct-2016 to 10-Nov-2016	-26.35%	6.54% / 9.50%	6.05% / 9.50% / 7.57%	41.35%	156.75%	37763.24	3.48x
2016 (27DEC2016 / Jan 2017)	UP	10-Nov-2016 to 07-Dec-2016	33.39%	8.35% / 13.85%	8.35% / 14.74% / 10.14%	39.52%	255.83%	24511.61	3.29x
2017 (23FEB2017 / Mar 2017)	DOWN	28-Dec-2016 to 17-Feb-2017	-25.81%	10.71% / 12.12%	0.03% / 16.30% / 13.41%	41.46%	3325.08%	24802.14	1.92x
2017 (25JAN2017 / Feb 2017)	DOWN	28-Dec-2016 to 09-Jan-2017	-18.75%	11.81% / 16.76%	11.25% / 16.76% / 14.16%	53.32%	-18.54%	59367.22	1.32x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2017 (28MAR2017 / Apr 2017)	DOWN	27-Jan-2017 to 27-Feb-2017	-22.21%	12.10% / 11.91%	0.02% / 12.98% / 11.43%	30.83%	2889.20%	8358.00	1.94x
2017 (25APR2017 / May 2017)	UP	27-Feb-2017 to 06-Apr-2017	15.37%	10.74% / 9.68%	9.12% / 12.28% / 10.76%	31.02%	1872.64%	15383.24	1.43x
2017 (27JUN2017 / Jul 2017)	DOWN	12-May-2017 to 21-Jun-2017	-17.07%	8.76% / 10.23%	8.36% / 10.85% / 9.42%	29.55%	115.86%	34258.10	1.81x
2017 (26DEC2017 / Jan 2018)	DOWN	13-Nov-2017 to 15-Dec-2017	-23.24%	7.43% / 12.46%	7.43% / 12.46% / 9.86%	35.13%	1860.29%	42578.80	2.36x
2018 (25JAN2018 / Feb 2018)	DOWN	29-Nov-2017 to 21-Dec-2017	-18.60%	9.40% / 13.28%	9.40% / 13.59% / 11.43%	40.32%	913.60%	7136.71	1.63x
2018 (25JAN2018 / Feb 2018)	UP	21-Dec-2017 to 17-Jan-2018	24.28%	13.28% / 15.41%	12.60% / 15.53% / 14.19%	44.99%	-12.92%	88587.47	1.71x
2018 (23FEB2018 / Mar 2018)	UP	26-Dec-2017 to 30-Jan-2018	21.19%	12.20% / 11.58%	11.48% / 14.34% / 12.57%	34.84%	85.79%	16460.60	1.69x
2018 (23FEB2018 / Mar 2018)	DOWN	30-Jan-2018 to 15-Feb-2018	-20.81%	11.58% / 14.17%	11.58% / 16.08% / 14.96%	45.61%	175.78%	79313.85	1.39x
2018 (26OCT2018 / Nov 2018)	UP	14-Sep-2018 to 10-Oct-2018	21.69%	5.14% / 11.07%	5.14% / 11.75% / 8.63%	36.75%	307.97%	28292.61	2.51x
2018 (27NOV2018 / Dec 2018)	UP	28-Sep-2018 to 14-Nov-2018	54.53%	7.68% / 27.90%	7.68% / 27.90% / 10.59%	64.81%	440.19%	36752.42	5.15x
2018 (26DEC2018 / Jan 2019)	UP	29-Oct-2018 to 14-Nov-2018	46.95%	8.37% / 27.63%	8.08% / 27.63% / 11.60%	86.44%	347.18%	5265.31	4.05x
2018 (26DEC2018 / Jan 2019)	DOWN	14-Nov-2018 to 17-Dec-2018	-27.97%	27.63% / 30.84%	27.63% / 39.41% / 33.17%	116.06%	-12.29%	48068.25	0.84x
2019 (28JAN2019 / Feb 2019)	DOWN	10-Dec-2018 to 07-Jan-2019	-36.52%	29.07% / 24.46%	24.46% / 30.56% / 27.70%	58.09%	958.20%	22459.75	1.32x
2019 (25FEB2019 / Mar 2019)	UP	03-Jan-2019 to 15-Jan-2019	16.69%	24.81% / 23.68%	20.87% / 24.81% / 23.05%	55.38%	40.98%	3220.89	0.72x
2019 (28JAN2019 / Feb 2019)	UP	07-Jan-2019 to 15-Jan-2019	22.07%	24.46% / 27.32%	22.11% / 28.22% / 24.58%	81.03%	-32.75%	60322.29	0.90x
2019 (25FEB2019 / Mar 2019)	DOWN	15-Jan-2019 to 07-Feb-2019	-21.07%	23.68% / 17.60%	16.68% / 23.68% / 19.79%	33.08%	925.32%	22357.78	1.06x
2019 (25APR2019 / May 2019)	DOWN	01-Mar-2019 to 18-Apr-2019	-15.15%	9.11% / 7.01%	6.62% / 9.11% / 7.52%	18.94%	2112.88%	12743.14	2.01x
2019 (25JUN2019 / Jul 2019)	DOWN	14-May-2019 to 20-Jun-2019	-20.25%	6.32% / 12.04%	5.87% / 12.04% / 8.54%	33.03%	1864.58%	24715.00	2.37x
2019 (26JUL2019 / Aug 2019)	DOWN	29-May-2019 to 20-Jun-2019	-19.14%	6.66% / 12.14%	6.66% / 12.14% / 9.55%	38.66%	3872.81%	2402.12	2.00x
2019 (25SEP2019 / Oct 2019)	UP	29-Jul-2019 to 17-Sep-2019	28.76%	12.01% / 10.01%	9.81% / 12.80% / 10.97%	30.11%	319.93%	29518.92	2.62x
2019 (25OCT2019 / Nov 2019)	UP	27-Aug-2019 to 17-Sep-2019	19.96%	9.44% / 9.43%	9.18% / 10.05% / 9.60%	25.11%	163.25%	3032.12	2.08x
2019 (25OCT2019 / Nov 2019)	DOWN	17-Sep-2019 to 11-Oct-2019	-19.05%	9.43% / 9.11%	9.02% / 10.23% / 9.66%	21.39%	624.99%	56083.83	1.97x
2019 (25NOV2019 / Dec 2019)	UP	23-Oct-2019 to 05-Nov-2019	17.46%	8.05% / 11.73%	7.57% / 11.96% / 9.84%	39.48%	43.07%	98399.20	1.77x
2019 (26DEC2019 / Jan 2020)	DOWN	05-Nov-2019 to 09-Dec-2019	-25.10%	10.32% / 18.42%	9.85% / 18.42% / 12.81%	50.18%	847.71%	54855.12	1.96x
2020 (28JAN2020 / Feb 2020)	DOWN	03-Dec-2019 to 21-Jan-2020	-24.31%	14.94% / 15.39%	11.70% / 16.73% / 14.12%	44.80%	317.02%	75671.37	1.72x
2020 (25FEB2020 / Mar 2020)	DOWN	07-Jan-2020 to 10-Feb-2020	-19.27%	10.46% / 12.75%	10.01% / 13.19% / 11.53%	37.12%	342.53%	60105.76	1.67x
2020 (26MAR2020 / Apr 2020)	UP	28-Feb-2020 to 11-Mar-2020	17.46%	13.35% / 16.30%	13.08% / 16.41% / 14.17%	49.12%	-48.91%	167907.00	1.23x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2020 (27APR2020 / May 2020)	UP	28-Feb-2020 to 11-Mar-2020	16.65%	12.05% / 15.33%	11.91% / 15.39% / 12.97%	47.77%	24.64%	6470.22	1.28x
2020 (26MAR2020 / Apr 2020)	DOWN	11-Mar-2020 to 18-Mar-2020	-18.42%	16.30% / 20.55%	16.30% / 20.55% / 17.28%	61.29%	4.67%	165090.17	1.07x
2020 (27APR2020 / May 2020)	DOWN	11-Mar-2020 to 03-Apr-2020	-19.01%	15.33% / 17.77%	15.33% / 19.27% / 17.09%	54.30%	611.29%	30041.12	1.11x
2020 (26MAY2020 / Jun 2020)	UP	03-Apr-2020 to 05-May-2020	21.55%	16.10% / 25.90%	16.10% / 28.38% / 23.16%	71.31%	1060.15%	62067.10	0.93x
2020 (27APR2020 / May 2020)	UP	03-Apr-2020 to 21-Apr-2020	21.58%	17.77% / 32.60%	17.77% / 32.60% / 24.28%	101.13%	-77.06%	64987.40	0.89x
2020 (25JUN2020 / Jul 2020)	DOWN	05-May-2020 to 17-Jun-2020	-29.78%	20.62% / 18.66%	18.11% / 25.50% / 21.64%	49.85%	1040.02%	82834.53	1.38x
2020 (26MAY2020 / Jun 2020)	DOWN	05-May-2020 to 13-May-2020	-23.16%	25.90% / 28.43%	25.90% / 28.43% / 26.91%	68.16%	94.41%	176787.29	0.86x
2020 (28JUL2020 / Aug 2020)	DOWN	01-Jun-2020 to 25-Jun-2020	-19.27%	19.00% / 20.16%	14.85% / 20.16% / 16.75%	38.80%	3344.49%	12134.95	1.15x
2020 (28JUL2020 / Aug 2020)	UP	25-Jun-2020 to 07-Jul-2020	20.90%	20.16% / 23.05%	19.52% / 24.58% / 22.28%	77.11%	-63.91%	186296.89	0.94x
2020 (26AUG2020 / Sep 2020)	UP	20-Jul-2020 to 19-Aug-2020	43.30%	18.75% / 27.69%	17.73% / 33.25% / 25.28%	75.54%	-33.39%	115404.17	1.71x
2020 (25SEP2020 / Oct 2020)	UP	27-Jul-2020 to 27-Aug-2020	36.09%	17.18% / 21.50%	17.03% / 28.35% / 23.19%	61.81%	877.24%	15789.67	1.56x
2020 (25SEP2020 / Oct 2020)	DOWN	27-Aug-2020 to 18-Sep-2020	-24.41%	21.50% / 23.93%	18.08% / 24.73% / 20.14%	49.42%	93.84%	233591.82	1.21x
2020 (27OCT2020 / Nov 2020)	DOWN	07-Sep-2020 to 01-Oct-2020	-17.04%	12.80% / 21.60%	12.33% / 22.70% / 15.88%	53.44%	114.07%	67491.79	1.07x
2020 (27OCT2020 / Nov 2020)	UP	01-Oct-2020 to 21-Oct-2020	22.07%	21.60% / 24.82%	21.60% / 24.82% / 23.53%	61.76%	15.44%	245749.50	0.94x
2020 (24NOV2020 / Dec 2020)	DOWN	30-Oct-2020 to 19-Nov-2020	-21.24%	11.19% / 18.75%	11.19% / 19.04% / 14.87%	46.90%	-14.99%	238127.87	1.43x
2020 (28DEC2020 / Jan 2021)	DOWN	30-Oct-2020 to 08-Dec-2020	-30.27%	10.02% / 23.75%	10.02% / 24.55% / 15.53%	50.79%	632.19%	84537.25	1.95x
2020 (28DEC2020 / Jan 2021)	UP	08-Dec-2020 to 22-Dec-2020	15.40%	23.75% / 19.47%	19.25% / 23.75% / 21.52%	24.85%	-48.12%	186569.36	0.72x
2021 (25JAN2021 / Feb 2021)	DOWN	30-Nov-2020 to 28-Dec-2020	-20.63%	14.29% / 24.57%	14.00% / 24.57% / 19.97%	61.71%	1288.36%	14648.55	1.03x
2021 (23FEB2021 / Mar 2021)	UP	28-Dec-2020 to 17-Feb-2021	36.72%	23.41% / 21.50%	17.89% / 24.23% / 21.27%	53.78%	1569.67%	117309.65	1.73x
2021 (25JAN2021 / Feb 2021)	UP	28-Dec-2020 to 12-Jan-2021	20.75%	24.57% / 22.14%	21.22% / 25.36% / 23.48%	59.83%	-46.53%	190892.67	0.88x
2021 (26MAR2021 / Apr 2021)	UP	25-Jan-2021 to 17-Feb-2021	15.82%	16.53% / 16.74%	16.53% / 20.27% / 18.39%	44.03%	211.45%	8401.88	0.86x
2021 (26MAR2021 / Apr 2021)	DOWN	17-Feb-2021 to 18-Mar-2021	-17.40%	16.74% / 13.64%	12.72% / 16.74% / 14.49%	27.67%	127.26%	124381.45	1.20x
2021 (25MAY2021 / Jun 2021)	UP	05-Apr-2021 to 17-May-2021	22.23%	12.44% / 13.24%	10.00% / 13.24% / 11.12%	31.24%	1053.24%	89532.74	2.00x
2021 (27JUL2021 / Aug 2021)	UP	28-May-2021 to 22-Jul-2021	34.73%	11.42% / 11.67%	11.14% / 13.98% / 12.26%	28.78%	483.18%	91771.30	2.83x
2021 (26AUG2021 / Sep 2021)	UP	07-Jul-2021 to 06-Aug-2021	16.33%	12.13% / 12.16%	10.73% / 12.93% / 11.75%	28.39%	691.98%	93483.65	1.39x
2021 (27SEP2021 / Oct 2021)	UP	19-Aug-2021 to 15-Sep-2021	40.03%	12.05% / 19.41%	10.90% / 20.16% / 16.00%	46.24%	60.60%	127798.40	2.50x
2021 (26OCT2021 / Nov 2021)	UP	30-Aug-2021 to 05-Oct-2021	47.64%	14.48% / 27.39%	14.07% / 27.39% / 19.66%	62.37%	96.39%	62751.78	2.42x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2021 (24NOV2021 / Dec 2021)	UP	29-Sep-2021 to 05-Oct-2021	15.82%	23.38% / 26.41%	23.33% / 26.41% / 24.13%	77.45%	34.24%	8059.00	0.66x
2021 (24NOV2021 / Dec 2021)	DOWN	05-Oct-2021 to 12-Nov-2021	-25.37%	26.41% / 31.24%	24.26% / 31.90% / 28.26%	76.44%	141.21%	47831.97	0.90x
2021 (26OCT2021 / Nov 2021)	DOWN	05-Oct-2021 to 19-Oct-2021	-19.69%	27.39% / 28.34%	26.93% / 30.69% / 28.41%	77.74%	-33.10%	137763.55	0.69x
2021 (28DEC2021 / Jan 2022)	DOWN	04-Nov-2021 to 06-Dec-2021	-36.60%	27.89% / 39.30%	26.80% / 39.30% / 31.06%	83.12%	688.27%	41350.35	1.18x
2022 (25JAN2022 / Feb 2022)	DOWN	26-Nov-2021 to 30-Dec-2021	-30.89%	28.41% / 29.36%	26.83% / 37.84% / 31.20%	72.35%	2603.22%	23475.88	0.99x
2022 (23FEB2022 / Mar 2022)	UP	30-Dec-2021 to 02-Feb-2022	53.43%	26.56% / 38.44%	21.01% / 38.44% / 25.03%	83.15%	313.12%	36270.00	2.13x
2022 (25JAN2022 / Feb 2022)	UP	30-Dec-2021 to 12-Jan-2022	26.62%	29.36% / 30.95%	23.68% / 30.95% / 26.63%	77.37%	-5.23%	101459.50	1.00x
2022 (25JAN2022 / Feb 2022)	DOWN	12-Jan-2022 to 20-Jan-2022	-17.60%	30.95% / 30.95%	30.06% / 33.12% / 31.23%	104.02%	-76.55%	82486.57	0.56x
2022 (28MAR2022 / Apr 2022)	UP	27-Jan-2022 to 02-Feb-2022	17.98%	23.16% / 28.67%	23.16% / 28.67% / 26.08%	77.27%	60.10%	10941.80	0.69x
2022 (23FEB2022 / Mar 2022)	DOWN	02-Feb-2022 to 11-Feb-2022	-27.78%	38.44% / 40.32%	38.44% / 44.39% / 41.62%	126.06%	-29.62%	73778.88	0.67x
2022 (28MAR2022 / Apr 2022)	DOWN	02-Feb-2022 to 11-Feb-2022	-20.81%	28.67% / 29.77%	28.67% / 31.98% / 30.21%	83.54%	56.36%	11687.88	0.69x
2022 (28MAR2022 / Apr 2022)	UP	11-Feb-2022 to 23-Mar-2022	34.37%	29.77% / 23.61%	23.30% / 30.33% / 27.23%	55.05%	48.94%	56900.79	1.26x
2022 (26APR2022 / May 2022)	UP	28-Feb-2022 to 18-Apr-2022	76.35%	27.39% / 24.27%	20.23% / 27.39% / 23.40%	46.73%	918.84%	46019.20	3.26x
2022 (25MAY2022 / Jun 2022)	UP	29-Mar-2022 to 05-May-2022	61.06%	21.11% / 27.30%	19.84% / 28.47% / 23.75%	60.82%	621.73%	40348.26	2.57x
2022 (27JUN2022 / Jul 2022)	UP	28-Apr-2022 to 06-Jun-2022	33.25%	26.79% / 26.80%	23.73% / 32.21% / 27.26%	65.38%	960.38%	35628.18	1.22x
2022 (25MAY2022 / Jun 2022)	UP	10-May-2022 to 19-May-2022	15.84%	31.22% / 27.18%	27.18% / 31.26% / 29.40%	32.83%	0.14%	89872.25	0.54x
2022 (26JUL2022 / Aug 2022)	DOWN	06-Jun-2022 to 05-Jul-2022	-40.29%	25.27% / 36.99%	23.05% / 38.59% / 31.14%	86.67%	286.61%	34190.32	1.29x
2022 (27JUN2022 / Jul 2022)	DOWN	06-Jun-2022 to 20-Jun-2022	-28.07%	26.80% / 34.26%	24.78% / 35.72% / 29.89%	90.39%	-53.30%	85038.18	0.94x
2022 (26AUG2022 / Sep 2022)	DOWN	28-Jun-2022 to 05-Jul-2022	-18.38%	32.74% / 37.70%	31.97% / 39.39% / 36.17%	96.54%	21.01%	4138.33	0.51x
2022 (26AUG2022 / Sep 2022)	UP	05-Jul-2022 to 19-Aug-2022	71.56%	37.70% / 27.22%	27.22% / 40.16% / 33.67%	67.56%	771.88%	35318.42	2.13x
2022 (26JUL2022 / Aug 2022)	UP	05-Jul-2022 to 21-Jul-2022	47.61%	36.99% / 36.07%	35.82% / 39.88% / 37.55%	80.77%	-1.73%	71933.31	1.27x
2022 (27SEP2022 / Oct 2022)	UP	08-Aug-2022 to 22-Aug-2022	26.93%	26.61% / 25.14%	25.13% / 28.28% / 26.49%	56.04%	635.82%	10538.90	1.02x
2022 (27SEP2022 / Oct 2022)	DOWN	22-Aug-2022 to 22-Sep-2022	-23.03%	25.14% / 24.63%	20.86% / 26.46% / 23.38%	54.66%	2.03%	55022.83	0.98x
2022 (26OCT2022 / Nov 2022)	DOWN	29-Aug-2022 to 21-Oct-2022	-45.67%	21.52% / 26.94%	20.27% / 26.94% / 22.75%	55.03%	384.63%	38116.72	2.01x
2022 (25NOV2022 / Dec 2022)	DOWN	06-Oct-2022 to 21-Oct-2022	-24.41%	21.20% / 22.86%	19.43% / 22.86% / 20.62%	41.05%	541.44%	12401.33	1.18x
2022 (25NOV2022 / Dec 2022)	UP	21-Oct-2022 to 07-Nov-2022	27.85%	22.86% / 36.47%	22.86% / 36.47% / 28.03%	104.67%	-21.52%	68427.67	0.99x
2022 (27DEC2022 / Jan 2023)	UP	28-Oct-2022 to 23-Nov-2022	26.92%	21.41% / 31.49%	21.41% / 35.43% / 30.30%	87.26%	728.32%	15142.95	0.89x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2022 (25NOV2022 / Dec 2022)	DOWN	07-Nov-2022 to 11-Nov-2022	-18.09%	36.47% / 39.78%	36.47% / 42.82% / 40.41%	131.45%	40.04%	72627.60	0.45x
2022 (27DEC2022 / Jan 2023)	DOWN	23-Nov-2022 to 22-Dec-2022	-31.57%	31.49% / 39.98%	27.45% / 41.99% / 33.41%	89.18%	29.70%	67930.55	0.94x
2023 (25JAN2023 / Feb 2023)	DOWN	29-Nov-2022 to 20-Jan-2023	-54.82%	27.83% / 38.54%	26.91% / 43.66% / 35.09%	84.27%	1419.38%	39372.00	1.56x
2023 (23FEB2023 / Mar 2023)	DOWN	26-Dec-2022 to 17-Feb-2023	-53.64%	29.10% / 28.90%	26.08% / 35.99% / 29.73%	63.33%	1017.82%	45426.62	1.80x
2023 (28MAR2023 / Apr 2023)	DOWN	27-Jan-2023 to 21-Feb-2023	-24.33%	24.85% / 24.80%	22.29% / 25.56% / 23.94%	51.30%	938.55%	14733.72	1.02x
2023 (28MAR2023 / Apr 2023)	UP	21-Feb-2023 to 03-Mar-2023	31.83%	24.80% / 28.06%	24.80% / 28.06% / 26.82%	66.71%	-32.98%	101790.78	1.19x
2023 (25APR2023 / May 2023)	DOWN	03-Mar-2023 to 13-Apr-2023	-35.34%	23.58% / 25.37%	23.58% / 30.54% / 26.27%	63.37%	1948.25%	52406.14	1.35x
2023 (28MAR2023 / Apr 2023)	DOWN	03-Mar-2023 to 23-Mar-2023	-27.04%	28.06% / 30.10%	28.06% / 37.08% / 32.77%	80.03%	6.67%	87589.20	0.83x
2023 (25MAY2023 / Jun 2023)	DOWN	18-Apr-2023 to 04-May-2023	-16.41%	23.70% / 22.00%	21.27% / 23.70% / 22.61%	45.13%	588.74%	88480.23	0.73x
2023 (25MAY2023 / Jun 2023)	UP	04-May-2023 to 19-May-2023	23.02%	22.00% / 21.90%	20.20% / 22.60% / 21.39%	46.20%	-62.35%	144959.58	1.08x
2023 (27JUN2023 / Jul 2023)	UP	04-May-2023 to 19-May-2023	17.75%	19.14% / 19.16%	17.89% / 19.77% / 18.72%	42.42%	157.34%	15744.00	0.95x
2023 (27JUN2023 / Jul 2023)	DOWN	19-May-2023 to 01-Jun-2023	-19.96%	19.16% / 20.52%	19.16% / 21.06% / 20.11%	38.86%	487.27%	96447.90	0.99x
2023 (26JUL2023 / Aug 2023)	UP	01-Jun-2023 to 26-Jun-2023	23.83%	18.77% / 19.84%	16.48% / 20.79% / 18.68%	48.18%	115.11%	22756.67	1.28x
2023 (27JUN2023 / Jul 2023)	UP	01-Jun-2023 to 19-Jun-2023	21.39%	20.52% / 20.84%	17.86% / 21.61% / 19.65%	51.13%	-65.00%	144480.15	1.09x
2023 (26SEP2023 / Oct 2023)	UP	02-Aug-2023 to 09-Aug-2023	16.71%	14.90% / 18.13%	14.83% / 18.13% / 16.10%	49.20%	10.60%	6302.00	1.04x
2023 (28AUG2023 / Sep 2023)	UP	02-Aug-2023 to 09-Aug-2023	18.59%	16.24% / 20.06%	16.24% / 20.06% / 17.95%	55.65%	-58.92%	90099.17	1.04x
2023 (26SEP2023 / Oct 2023)	DOWN	09-Aug-2023 to 06-Sep-2023	-15.94%	18.13% / 16.75%	16.42% / 19.65% / 17.93%	44.00%	1745.01%	60137.65	0.89x
2023 (28AUG2023 / Sep 2023)	DOWN	09-Aug-2023 to 23-Aug-2023	-15.06%	20.06% / 21.30%	20.06% / 22.76% / 21.55%	57.93%	35.73%	103556.80	0.70x
2023 (26OCT2023 / Nov 2023)	UP	26-Sep-2023 to 09-Oct-2023	18.64%	13.00% / 15.33%	13.00% / 15.50% / 13.74%	35.27%	-9.42%	110118.67	1.36x
2023 (27NOV2023 / Dec 2023)	DOWN	10-Oct-2023 to 21-Nov-2023	-21.43%	13.00% / 19.69%	13.00% / 20.92% / 16.50%	47.85%	518.45%	65753.94	1.30x
2023 (26DEC2023 / Jan 2024)	DOWN	31-Oct-2023 to 12-Dec-2023	-38.90%	16.94% / 21.41%	14.80% / 21.99% / 16.98%	46.46%	678.54%	42057.77	2.29x
2024 (25JAN2024 / Feb 2024)	DOWN	27-Nov-2023 to 12-Dec-2023	-21.58%	20.75% / 23.68%	18.75% / 24.33% / 20.71%	45.93%	157.01%	6759.00	1.04x
2024 (25JAN2024 / Feb 2024)	UP	12-Dec-2023 to 12-Jan-2024	43.91%	23.68% / 32.52%	19.16% / 33.18% / 23.41%	69.15%	16.61%	78700.61	1.88x
2024 (26FEB2024 / Mar 2024)	DOWN	09-Jan-2024 to 19-Feb-2024	-42.91%	21.05% / 21.23%	19.48% / 22.99% / 21.49%	49.31%	333.84%	58498.76	2.00x
2024 (25JAN2024 / Feb 2024)	DOWN	12-Jan-2024 to 19-Jan-2024	-23.03%	32.52% / 33.56%	32.52% / 33.68% / 33.11%	53.29%	-12.76%	110645.50	0.70x
2024 (25MAR2024 / Apr 2024)	DOWN	25-Jan-2024 to 19-Feb-2024	-25.50%	21.52% / 18.76%	18.39% / 21.75% / 19.83%	28.65%	584.42%	11258.71	1.29x
2024 (25MAR2024 / Apr 2024)	UP	19-Feb-2024 to 04-Mar-2024	17.66%	18.76% / 25.89%	18.18% / 26.37% / 24.45%	74.11%	49.27%	83504.00	0.72x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2024 (25APR2024 / May 2024)	DOWN	04-Mar-2024 to 16-Apr-2024	-19.45%	23.59% / 19.05%	16.60% / 23.59% / 19.54%	43.48%	1318.36%	58760.84	1.00x
2024 (25JUN2024 / Jul 2024)	UP	26-Apr-2024 to 11-Jun-2024	34.29%	17.85% / 21.39%	15.29% / 21.39% / 17.84%	46.91%	449.41%	54642.85	1.92x
2024 (28MAY2024 / Jun 2024)	UP	26-Apr-2024 to 22-May-2024	45.92%	14.32% / 20.01%	14.32% / 20.01% / 17.67%	46.65%	-61.91%	125372.11	2.60x
2024 (26JUL2024 / Aug 2024)	UP	31-May-2024 to 11-Jun-2024	17.68%	17.18% / 19.40%	17.18% / 19.40% / 17.92%	42.15%	12.45%	10269.50	0.99x
2024 (26JUL2024 / Aug 2024)	DOWN	11-Jun-2024 to 17-Jul-2024	-35.00%	19.40% / 21.50%	17.94% / 23.10% / 20.20%	41.75%	928.34%	63195.93	1.73x
2024 (27AUG2024 / Sep 2024)	DOWN	28-Jun-2024 to 05-Aug-2024	-24.62%	19.13% / 18.34%	16.30% / 20.44% / 18.59%	41.76%	696.09%	46974.48	1.32x
2024 (25SEP2024 / Oct 2024)	UP	05-Aug-2024 to 20-Sep-2024	16.49%	18.87% / 18.04%	16.37% / 19.45% / 18.20%	43.27%	307.49%	92642.18	0.91x
2024 (28OCT2024 / Nov 2024)	UP	30-Aug-2024 to 03-Oct-2024	18.27%	17.28% / 16.82%	14.13% / 18.08% / 16.05%	37.63%	417.37%	59307.33	1.14x
2024 (25NOV2024 / Dec 2024)	DOWN	03-Oct-2024 to 08-Nov-2024	-20.40%	20.24% / 18.66%	16.75% / 20.62% / 18.77%	41.86%	222.23%	50086.37	1.09x
2024 (28OCT2024 / Nov 2024)	DOWN	03-Oct-2024 to 18-Oct-2024	-23.39%	16.82% / 19.15%	16.82% / 19.43% / 18.05%	39.38%	114.48%	110153.25	1.30x
2024 (25NOV2024 / Dec 2024)	UP	08-Nov-2024 to 20-Nov-2024	19.22%	18.66% / 23.24%	18.66% / 23.98% / 22.52%	64.27%	-56.60%	108749.67	0.85x
2024 (26DEC2024 / Jan 2025)	UP	08-Nov-2024 to 20-Dec-2024	29.64%	16.56% / 26.58%	16.56% / 26.58% / 20.93%	61.37%	357.50%	98363.74	1.42x
2025 (28JAN2025 / Feb 2025)	UP	04-Dec-2024 to 16-Jan-2025	43.03%	27.14% / 32.44%	24.26% / 38.48% / 30.19%	78.23%	232.50%	97836.90	1.43x
2025 (25FEB2025 / Mar 2025)	UP	27-Dec-2024 to 19-Feb-2025	46.93%	23.68% / 26.58%	19.73% / 28.51% / 23.64%	59.96%	410.11%	58386.54	1.99x
2025 (26MAR2025 / Apr 2025)	UP	31-Jan-2025 to 10-Mar-2025	47.03%	20.62% / 26.43%	18.92% / 26.43% / 22.69%	61.11%	743.33%	60603.37	2.07x
2025 (25APR2025 / May 2025)	UP	28-Feb-2025 to 10-Mar-2025	16.12%	30.28% / 31.19%	30.08% / 32.67% / 31.00%	72.25%	142.57%	16174.14	0.52x
2025 (25APR2025 / May 2025)	DOWN	10-Mar-2025 to 16-Apr-2025	-30.54%	31.19% / 24.45%	21.86% / 31.56% / 26.03%	52.84%	517.22%	68447.96	1.17x
2025 (27MAY2025 / Jun 2025)	DOWN	03-Apr-2025 to 24-Apr-2025	-27.69%	29.27% / 27.01%	27.01% / 30.87% / 29.08%	52.62%	433.64%	18369.73	0.95x
2025 (27MAY2025 / Jun 2025)	UP	24-Apr-2025 to 09-May-2025	22.35%	27.01% / 25.43%	24.89% / 27.77% / 26.20%	54.32%	33.91%	97217.33	0.85x
2025 (27MAY2025 / Jun 2025)	DOWN	09-May-2025 to 19-May-2025	-16.90%	25.43% / 25.01%	23.55% / 25.43% / 24.65%	52.46%	25.53%	101444.29	0.69x
2025 (25JUN2025 / Jul 2025)	UP	30-May-2025 to 19-Jun-2025	18.99%	23.58% / 21.28%	20.88% / 24.83% / 22.30%	44.08%	10.80%	98665.87	0.85x
2025 (28JUL2025 / Aug 2025)	UP	30-May-2025 to 19-Jun-2025	18.77%	23.30% / 20.40%	20.16% / 24.27% / 21.69%	40.58%	431.55%	13223.80	0.87x
2025 (28JUL2025 / Aug 2025)	DOWN	19-Jun-2025 to 23-Jul-2025	-26.01%	20.40% / 22.88%	20.40% / 25.40% / 22.69%	55.77%	40.65%	95936.00	1.15x
2025 (26AUG2025 / Sep 2025)	DOWN	27-Jun-2025 to 20-Aug-2025	-25.48%	24.74% / 19.62%	18.56% / 26.72% / 21.76%	48.22%	1140.47%	59048.95	1.17x
2025 (28OCT2025 / Nov 2025)	DOWN	01-Oct-2025 to 16-Oct-2025	-15.70%	13.59% / 15.40%	13.59% / 16.32% / 14.86%	41.64%	77.02%	157645.45	1.06x
2025 (24NOV2025 / Dec 2025)	UP	16-Oct-2025 to 13-Nov-2025	28.51%	13.00% / 17.41%	13.00% / 17.47% / 16.14%	44.84%	174.02%	107527.38	1.77x
2025 (28OCT2025 / Nov 2025)	UP	16-Oct-2025 to 22-Oct-2025	17.55%	15.40% / 21.94%	15.23% / 23.13% / 19.64%	71.62%	-63.63%	102882.80	0.89x

Year & Contract	Direction	Price Leg (Dates)	Price Move %	Margin % (Start / End)	Margin % (Min / Max / Avg)	Ann. Vol %	OI Change %	Avg Vol (Lots)	Leverage Eff.
2025 (26DEC2025 / Jan 2026)	UP	29-Oct-2025 to 05-Dec-2025	33.85%	13.07% / 19.95%	13.07% / 19.95% / 14.93%	43.90%	859.44%	76119.11	2.27x
2025 (26DEC2025 / Jan 2026)	DOWN	05-Dec-2025 to 16-Dec-2025	-27.05%	19.95% / 26.79%	19.95% / 28.51% / 25.33%	85.69%	-47.56%	135632.38	1.07x
2026 (27JAN2026 / Feb 2026)	DOWN	05-Dec-2025 to 16-Jan-2026	-37.44%	17.77% / 24.98%	17.77% / 26.60% / 23.46%	63.15%	345.61%	87647.73	1.60x
2026 (24FEB2026 / Mar 2026)	DOWN	29-Dec-2025 to 14-Jan-2026	-20.48%	18.93% / 24.02%	18.93% / 24.02% / 21.57%	54.12%	95.18%	19281.38	0.95x
2026 (24FEB2026 / Mar 2026)	UP	14-Jan-2026 to 30-Jan-2026	60.90%	24.02% / 34.86%	22.53% / 34.86% / 29.70%	88.44%	-36.82%	106982.17	2.05x
2026 (27JAN2026 / Feb 2026)	UP	16-Jan-2026 to 22-Jan-2026	65.23%	24.98% / 52.71%	24.98% / 53.65% / 41.29%	152.11%	-61.89%	192153.60	1.58x
2026 (24FEB2026 / Mar 2026)	DOWN	30-Jan-2026 to 18-Feb-2026	-31.14%	34.86% / 47.49%	34.86% / 66.05% / 54.90%	132.91%	18.75%	80718.21	0.57x
2026 (26MAR2026 / Apr 2026)	DOWN	30-Jan-2026 to 26-Feb-2026	-28.66%	28.05% / 33.10%	28.05% / 54.84% / 42.25%	95.04%	955.42%	23333.35	0.68x
2026 (26MAR2026 / Apr 2026)	UP	26-Feb-2026 to 12-Mar-2026	15.75%	33.10% / 32.33%	31.44% / 33.43% / 32.49%	71.68%	-26.08%	103629.45	0.48x
2026 (27APR2026 / May 2026)	DOWN	12-Mar-2026 to 14-Apr-2026	-18.74%	31.48% / 22.04%	22.04% / 31.48% / 26.11%	40.31%	668.01%	53112.57	0.72x
2026 (26MAY2026 / Jun 2026)	DOWN	27-Mar-2026 to 29-Apr-2026	-17.05%	20.73% / 15.01%	14.90% / 21.87% / 18.16%	32.72%	2131.18%	22969.78	0.94x
2026 (26MAY2026 / Jun 2026)	UP	29-Apr-2026 to 19-May-2026	18.76%	15.01% / 16.84%	14.85% / 17.94% / 16.44%	43.90%	-30.11%	104730.47	1.14x

TASK 3 - MARGIN vs PRICE CORRELATION

1. Lead / Lag / Simultaneous Classifications

1. Lead / Lag / Simultaneous Classifications

1. Lead / Lag / Simultaneous Classifications

Across **213 contract swings** analyzed since 2010 under the strict 5 <= DTE <= 60 tradeable window (using a forward-fill for dates with missing margins):

Classification	Description	Count	Percentage
Leads	Peak margin occurred before the price swing start.	35	16.43%
Simultaneous	Peak margin occurred during the price swing.	94	44.13%
Lags	Peak margin occurred after the price swing completed.	84	39.44%

Interpretation:

- Lagging peak margin is typical for major completed up moves, with peak margin lagging the price peak by **3 to 9 trading days** (as historical volatility parameters update in the SPAN model).

- Simultaneous peak margin occurs when high volatility expands during the price move, triggering intraday or daily margin revisions before the price move is finished.

- Leading peak margin occurs when the exchange maintains a defensive high margin due to historical seasonality or recent macro events before the contract begins its main directional move.

2. Return vs Margin Change Correlation

Over the last 2 years, the correlation between the **5-day price return at t** and the **next-day margin change (percentage points) at $t+1$** is **-0.11**. This confirms a very weak linear relationship on a day-to-day basis. Margin updates are structural, driven by multi-week volatility regimes rather than short-term returns.

3. Preemptive Margin Spikes (Quiet Price Action)

We identified **23 instances** since 2010 where a contract's initial margin spiked by **> 2.00 pp** in a single session while the absolute 5-day price return remained quiet (**< 3.00%**). Below are the most notable spikes:

Spike Date	Contract Expiry	Margin Change	5-Day Price Return %	Close Price
20-May-2020 (Jun 2020)	25MAY2020 (Jun 2020)	+4.03 pp	+1.16%	139.50
09-May-2022 (Jun 2022)	25MAY2022 (Jun 2022)	+3.67 pp	-1.48%	564.30

Spike Date	Contract Expiry	Margin Change	5-Day Price Return %	Close Price
28-Oct-2024 (Nov 2024)	28OCT2024 (Nov 2024)	+5.35 pp	-0.36%	193.30
30-Dec-2025 (Jan 2026)	26DEC2025 (Jan 2026)	+3.83 pp	+1.91%	357.90
02-Feb-2026 (Mar 2026)	24FEB2026 (Mar 2026)	+31.19 pp	+2.02%	296.80

Implication: These spikes signal the clearing corporation's preemptive risk mitigation. It occurs when implied volatility surges or during rolling windows before high-impact events, even if price remains temporarily rangebound.

TASK 4 - CURRENT MARKET STATE ASSESSMENT

- **Current Margin Regime**: **Medium**. The current initial margin of **15.84%** resides in the **64.66th percentile** of historical data. The extreme volatility of Q1 2026 has fully abated, pulling margins down from the **66.05%** stress peak.
- **Recent Price Regime**: **Pullback / Consolidation**. Following a major swing low on **24-Apr-2026** (238.90), the contract rallied **+32.27%** to peak at **316.00** on **29-May-2026**, and is currently consolidating around **302.40**.
- **Most Recent Major Move**: **UP** move from **24-Apr-2026** to **29-May-2026** (**+32.27%** over 26 trading days). The price action has broken its swing high and is currently resting.
- **Trader Watchlist (Key Patterns)**:
- **Margin-Price Divergence**: During the +32.27% price rally, margin requirements actually fell from **18.51%** to **15.48%**. This divergence indicates margins are currently underpriced relative to the recent price trend.
- Directional traders should prepare for a potential margin hike if prices resume their uptrend above the **316.00** level.

DATA QUALITY AND DATABASE FRESHNESS

- **Data Quality**: The datasets contain zero duplicate records in `margins.db` or the `FUTCOM` portion of `prices.db`. There are no missing values in the primary analytical columns.
- **Database Date Ranges**:
- `data/margins.db`: **01-Jan-2010** to **03-Jun-2026**
- `data/prices.db`: **01-Jan-2007** to **02-Jun-2026**